Multi-Objective Online Learning

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References

Chapter 3 of Aaron's lecture notes:
https://www.cis.upenn.edu/~aaroth/GamesInLearning.pdf

Online Learning with Multiple Subsequences

For $t = 1, 2, \ldots, T$:

Learner observes a context $C_t \in \{0,1\}^K$

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \ldots, \ell_t(N)\right) \in \mathbb{R}^N$

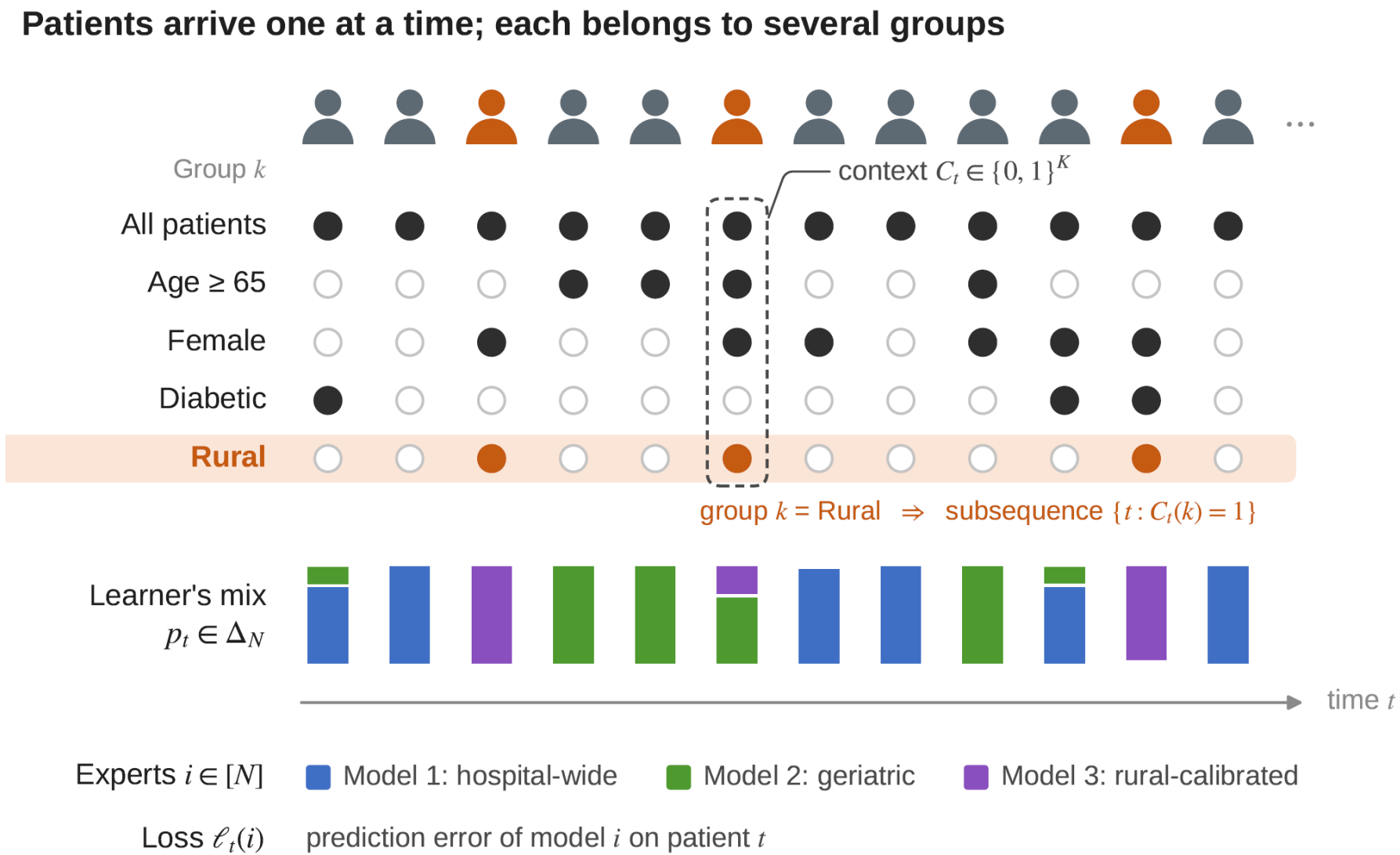
Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

$$\text{Group-wise Regret} = \max_{k \in [K]} \left(\sum_{t:\, C_t(k)=1} \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t:\, C_t(k)=1} \ell_t(i) \right)$$

$$= \max_{k \in [K]} \max_{i \in [N]} \sum_{t=1}^{T} C_t(k) \left(\langle p_t, \ell_t \rangle - \ell_t(i) \right)$$

Online Learning with Multiple Subsequences



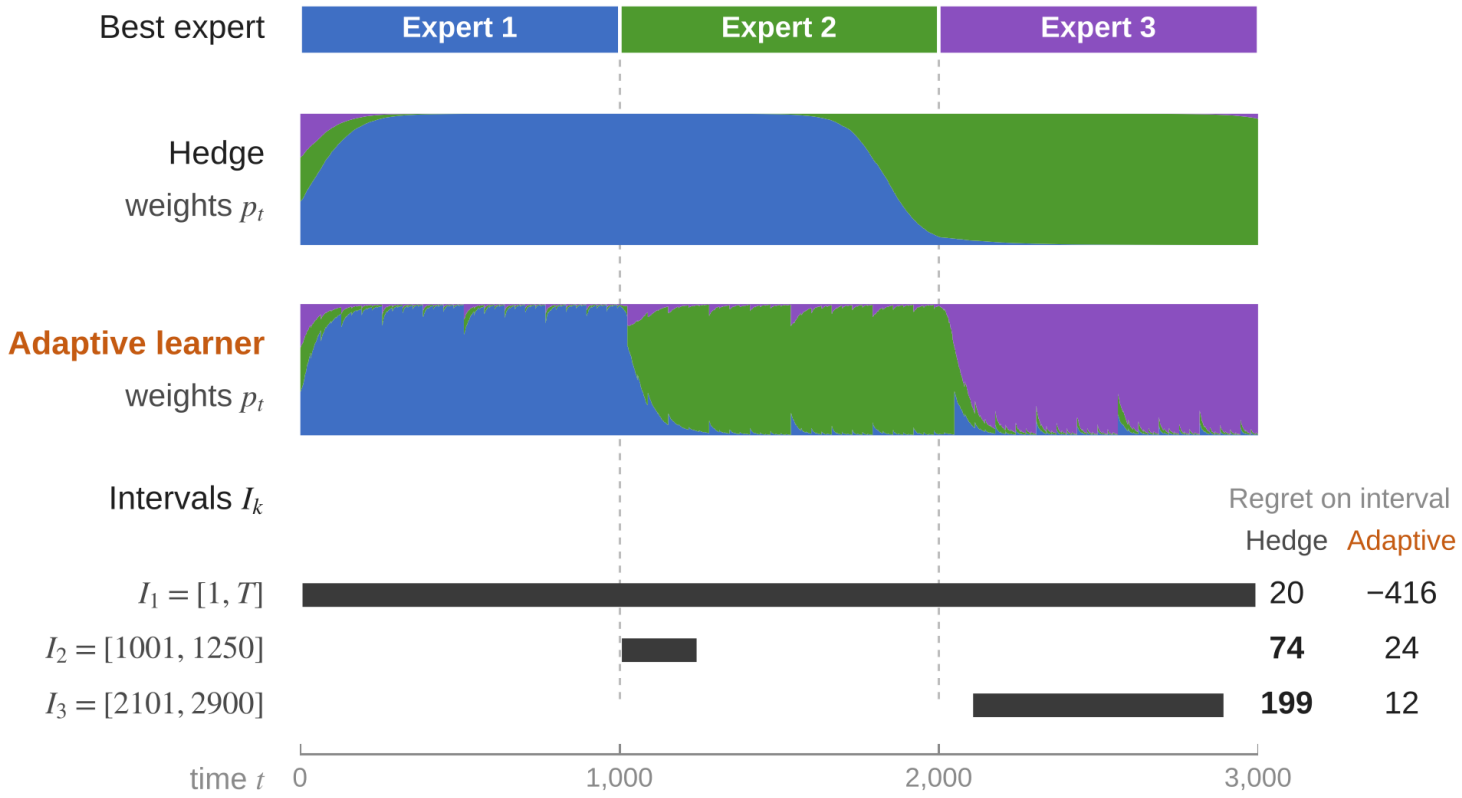
Why care about group-wise regret?

Online Learning with Multiple Subsequences

(a) The best expert changes over time; every interval is a group

Adaptive regret $= \max_{I=[s,e]\subseteq[T]} \left(\sum_{t\in I} \langle p_t, \ell_t \rangle - \min_{i\in[N]} \sum_{t\in I} \ell_t(i) \right)$

= group-wise regret with one group per interval: $C_t(k) = \mathbf{1}\{t \in I_k\}$



Is the standard no-regret guarantee sufficient?

No.

Consider loss sequence $\ell_t = (0,1)$ for $t \le \frac{T}{2}$

$\ell_t = (1,0)$ for $t \in [\frac{T}{2}, T]$

Since $L_t(1) \le L_t(2)$ on any t, if we run exponential weights on it, we have $p_t(1) \ge p_t(2)$ on any t

This makes $\sum_{t=T/2}^{T} \langle p_t, \ell_t \rangle - \sum_{t=T/2}^{T} \ell_t(2) = \Omega(T)$

Our goal

$$\max_{k\in[K]} \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(i)) \leq O\left(\sqrt{T \log(NK)}\right)$$

In fact, the following stronger guarantee is possible:

$$\forall k \in [K] \qquad \max_{i\in[N]} \sum_{t=1}^{T} C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(i)) \leq O\left(\sqrt{\left(\sum_{t=1}^{T} C_t(k)\right) \log(NK)}\right)$$

Recall: No-Regret Best-Response Dynamics

With a known $G \in \mathbb{R}^{M \times N}$

p-player (minimizer)

For $t = 1,2, \ldots$

Choose $p_t = \underset{p \in \Delta_M}{\operatorname{argmin}}\, p^\top G q_t$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $G^\top p_t \in \mathbb{R}^N$

Boosting:	Model player	Sample player
LP:	Variable player	Constraint player

Recall: No-Regret Best-Response Dynamics

Guarantee:

$$\max_{q} \sum_{t=1}^{T} p_t^\top G q \le T \cdot \text{GameValue}(G) + O\left(\sqrt{T \log N}\right)$$

$\min_p \max_q p^\top G q$

By the regret guarantee of the q-player,

$$\max_{q} \sum_{t=1}^{T} p_t^\top G q - \sum_{t=1}^{T} p_t^\top G q_t \le O\left(\sqrt{T \log N}\right)$$

$$p_t^\top G q_t = \min_p p^\top G q_t \le \min_p \max_q p^\top G q$$

Modified Scenario

With a time-varying and unknown $G_t \in \mathbb{R}^{M \times N}$

Additional assumption:

$$\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \leq V^\star$$

p-player (minimizer)

For $t = 1,2, \ldots$

~~Choose $p_t = \underset{p \in \Delta_M}{\operatorname{argmin}} \, p^\top G q_t$~~

Choose p_t such that $p_t^\top G_t q_t \leq V^\star$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $G_t^\top p_t \in \mathbb{R}^N$

Modified Scenario

Guarantee:

$$\max_{q} \sum_{t=1}^{T} p_t^\top G_t q \leq TV^\star + O\left(\sqrt{T \log N}\right)$$

By the regret guarantee of the $\{p_t\}$-player.

$$\max_{\{q_t\}} \sum_{t=1}^{T} p_t^\top G_t q_t - \sum_{t=1}^{T} p_t^\top G_t q_t \leq O\left(\sqrt{T \log N}\right)$$

$$\leq V^*$$

A “trivial” instantiation of the dynamics

$G_t \in \mathbb{R}^{N \times N}$

$G_t(i,j) = \ell_t(i) - \ell_t(j)$

Verify: $\forall q \ \exists p \ \forall G_t \quad p^\top G_t q \le 0$ ✓*

$$p^\top G_t q = \sum_{i,j} p(i)\, q(j)\, G_t(i,j)$$

$$= \sum_{i,j} p(i)\, q(j) \left(\ell_t(i) - \ell_t(j) \right)$$

$$= \langle p, \ell_t \rangle - \langle q, \ell_t \rangle$$

Given q, just set $p = q$.

Guarantee:

$$\max_q \sum_{t=1}^{T} p_t^\top G_t q \le O\left(\sqrt{T \log N}\right)$$

$$\Rightarrow \max_q \sum_{t=1}^{T} \sum_{i,j} p_t(i)\, q(j) \left(\ell_t(i) - \ell_t(j) \right) \le O\left(\sqrt{T \log N}\right)$$

$$\Rightarrow \max_q \sum_{t=1}^{T} \left(\langle p_t, \ell_t \rangle - \langle q, \ell_t \rangle \right) \le O\left(\sqrt{T \log N}\right)$$

p-player's regret over loss ℓ_t

A “trivial” instantiation of the dynamics

$G_t \in \mathbb{R}^{N \times N}$

$G_t(i,j) = \ell_t(i) - \ell_t(j)$

The interaction just looks like below (somewhat trivial):

p-player (minimizer)

For $t = 1,2, \ldots$

Choose $p_t = q_t$

q-player (maximizer)

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_N$ from no-regret alg

Feed no-regret alg with reward $r_t(j) = \langle p_t, \ell_t \rangle - \ell_t(j)$

Back to "Online Learning with Multiple Subsequences"

For $t = 1, 2, \dots, T$:

Learner observes a context $C_t \in \{0,1\}^K$

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss vector $\ell_t = \left(\ell_t(1), \dots, \ell_t(N)\right) \in \mathbb{R}^N$

Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

$$\text{Group-wise Regret} = \max_{k \in [K]} \left(\sum_{t:\, C_t(k)=1} \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t:\, C_t(k)=1} \ell_t(i) \right)$$

$$= \max_{k \in [K]} \max_{i \in [N]} \sum_{t=1}^{T} C_t(k) \left(\langle p_t, \ell_t \rangle - \ell_t(i) \right)$$

Back to “Online Learning with Multiple Subsequences”

$G_t \in \mathbb{R}^{N \times NK}$

$G_t(i,(j,k)) = C_t(k)(\ell_t(i) - \ell_t(j))$

Verify: $\forall q, C_t \ \exists p \ \forall \ell_t \qquad p^\top G_t q \leq 0$

Back to "Online Learning with Multiple Subsequences"

$G_t \in \mathbb{R}^{N \times NK}$

$G_t(i,(j,k)) = C_t(k)(\ell_t(i) - \ell_t(j))$

Guarantee:

$$\max_{\{j\}} \sum_{t=1}^{T} p_t^T G_t \{j\} \le O(\sqrt{T \log NK})$$

$$\Rightarrow \max_{\{j\}} \sum_{t=1}^{T} \sum_{i,j,k} p_t(i)\, g(j,k)\, C_t(k)\left(\ell_t(i) - \ell_t(j)\right) \le O(\sqrt{T \log(NK)})$$

$$\Rightarrow \max_{j,k} \sum_{t=1}^{T} \sum_{i} p_t(i)\, C_t(k)\left(\ell_t(i) - \ell_t(j)\right) \le O(\sqrt{T \log(NK)})$$

$$= \max_{k} \max_{j} \sum_{t=1}^{T} C_t(k)\left(\langle p_t, \ell_t \rangle - \ell_t(j)\right)$$ ← group-wise regret

Resulting Algorithm

For $t = 1,2, \ldots$

Choose $p_t \in \Delta_N$ as

$$p_t(i) = \frac{\sum_k q_t(i,k) C_t(k)}{\sum_{j,k} q_t(j,k) C_t(k)}$$

For $t = 1,2, \ldots$

Generate $q_t \in \Delta_{NK}$ from no-regret alg

Feed no-regret alg with reward

$$r_t(j,k) = C_t(k)(\langle p_t, \ell_t \rangle - \ell_t(j))$$